

Periodic Motion

Concepts, derivations, and why oscillations matter

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Outline

- ✓ Introduction
- ✓ What is Oscillation
- ✓ Harmonic Oscillation
- ☐ Terminologies in SHM
- ☐ Energy in SHM
- ☐ Linearity and Time-Translation Invariance
- ☐ The Complex Exponential and Rotating Phasors
- ☐ Solving the equation of motion of SHM

Learning Arc

We align the lecture with our study guide. We focus on conceptual definitions first, and then build the mathematical framework step-by-step.

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SECTION 1

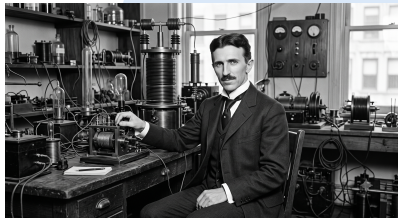
Introduction

A Moment of Pondering

Nikola Tesla

“If you want to find the secrets of the universe, think in terms of energy, frequency, and vibration.”

- Before we begin, close your eyes, cover your ears, sit in silence for a minute, and ponder this saying.
- What did you observe? A silent void, perhaps.
- But even that silence tells us something important.



Silence vs. Oscillation

- **Sound** depends on the oscillation and vibration of particles in a medium.
- **Light** depends on oscillating electric and magnetic fields.
- The universe would be a silent, uninteresting void without oscillations.

Fundamental Question

What makes oscillations so special? Why do so many physical phenomena depend on oscillation and wave motion?



Photo credit: Pexels

Why It Matters for Engineering

Useful Applications

Oscillations make many technologies possible:

- Sound systems and acoustics
- Sensors and instrumentation
- Rotating and vibrating machinery
- Telecommunications

Engineering Hazards

If left unchecked, vibrations can:

- Damage machinery
- Fatigue and weaken materials
- Cause structural failure (e.g., bridge collapses)

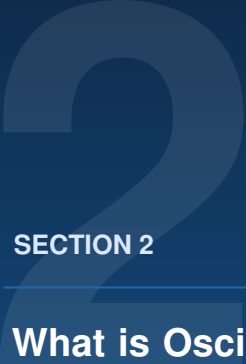
Our Quest in This Course

We begin our quest to obtain answers to these key questions:

1. What are the fundamental building blocks of vibrations?
2. What are the energy consequences of vibrations?
3. What happens when vibrations propagate from one place to another?
4. How can we employ, as well as control, vibrations to improve our lives?

The Language of Nature

By the end of this course, you should not see vibrations and waves as just formulas on a board. You should see them as one of the basic languages through which nature, machines, and engineering systems communicate.

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SECTION 2

What is Oscillation?

Periodic Motion vs. Oscillation

Definition 2.1 | Periodic Motion

Any motion that repeats itself after a specific time interval is called a **periodic motion**. Its position at a given time repeats itself after a fixed time interval, called the **period** T .

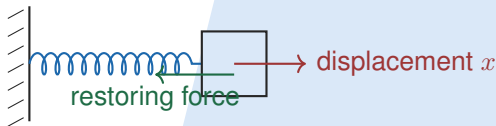
$$x(t) = x(t + T) \quad (1)$$

Definition 2.2 | Oscillation

An **oscillation** is a subset of periodic motion. It is the repeated variation of a physical quantity about an equilibrium or mean value.

Equilibrium and Restoring Force

- In a mechanical system, oscillation appears as the to-and-fro motion of a particle about its equilibrium position.
- **Equilibrium Position:** The position where the net force on the particle is zero.
- **Restoring Force:** A force that tends to bring the particle back toward equilibrium whenever it is displaced.
- If an oscillation occurs in a mechanical system, it is usually called a **vibration**.



Horizontal Mass-Spring

A mass on a frictionless surface is a classic mechanical oscillator.

The Mechanics of Overshooting

How does the oscillation sustain itself?

1. **Displacement:** Displace the mass from equilibrium. The spring stretches and exerts a restoring force towards equilibrium.
2. **Acceleration:** Released from rest, the mass accelerates toward the equilibrium position.
3. **Overshoot:** At equilibrium, the net force is zero, but the mass has gained velocity and inertia. It overshoots the equilibrium position.
4. **Compression:** The spring compresses, exerting a restoring force in the opposite direction, slowing the mass down until it stops and reverses.
5. **Ideal Oscillation:** Without friction or energy loss, this cycle continues indefinitely.

SECTION 3

Harmonic Oscillation

Hooke's Law

Definition 3.1 | Harmonic Oscillation

For an oscillation to be harmonic, the restoring force must be directly proportional to the displacement from equilibrium:

$$F = -K(x - x_0) \quad (2)$$

where K is the **force constant** and x_0 is the equilibrium position.

Hooke's Law

- This is a statement of **Hooke's Law**. A system obeying this is a **harmonic oscillator**.
- The minus sign indicates the restoring force acts in the opposite direction of displacement.
- **Idealization:** Hooke's law is an approximation, but works exceptionally well for sufficiently small displacements.

Derivation: Equation of Motion

Let us derive the differential equation for a simple harmonic oscillator. Set the coordinate origin at the equilibrium position ($x_0 = 0$):

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$$\implies ma = -Kx$$

Derivation: Equation of Motion

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Rearranging terms gives:

$$\boxed{m \frac{d^2}{dt^2}x(t) + Kx(t) = 0} \quad (3)$$

What is a “Simple” Harmonic Oscillator?

The equation we derived:

$$m \frac{d^2 x(t)}{dt^2} + Kx(t) = 0$$

Simple Harmonic Oscillator (SHO)

The word “**Simple**” implies that the system is:

- **Free:** No external forces acting on it.
- **Undamped:** No friction, air resistance, or mechanical energy loss.
- **Unforced:** No driving forces pushing the system.
- **Linear:** The restoring force is directly proportional to displacement.

General Solution

eq. (3) is a second-order, linear differential equation. The proposed general solution is:

Proposed General Solution

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t) \quad (4)$$

- c_1 and c_2 are arbitrary constants determined by the initial displacement and velocity of the particle.
- ω is the **angular frequency** of the oscillation, given by:

$$\omega = \sqrt{\frac{K}{m}}$$

Derivation: Verifying the Solution (Part 1)

To prove that our proposed $x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$ is a solution, we first compute its derivatives:

$$x'(t) = \frac{d}{dt} [c_1 \cos(\omega t) + c_2 \sin(\omega t)]$$

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Derivation: Verifying the Solution (Part 2)

Now, substitute $x''(t) = -\omega^2 x(t)$ back into the equation of motion:

$$m \frac{d^2 x(t)}{dt^2} + Kx(t) = 0$$

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This equation must hold for all times t . This is true if and only if the coefficient is zero:

$$K - m\omega^2 = 0 \implies \omega^2 = \frac{K}{m} \implies \omega = \sqrt{\frac{K}{m}}$$

Derivation: Releasing from Rest

For the unique case where the particle is released from rest at maximum displacement A :

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This yields the particular solution:

$$x(t) = A \cos(\omega t)$$

SECTION 4

Key Terminologies and Future Outlook

Key Terminologies

- **Amplitude (A):** The maximum displacement from the equilibrium position.
- **Future Note Sections to Come:**
 - Terminologies in SHM (Period, Frequency, Phase)
 - Energy in SHM (Kinetic and Potential Energy exchange)
 - Linearity and Time-Translation Invariance
 - Complex Exponential and Rotating Phasors ($e^{i\omega t}$)
 - Solving the equation of motion using complex numbers

Note Status

These subsequent topics will be incorporated as the lecture notes are drafted.

End of Lecture

PHY 104 – Periodic Motion